

Error Correction, Limits, and Honest Assessment

1, 3 December · Objective 8 · Final project presentations

Everything so far assumed the computation survives. It does not. This week covers how error correction rescues it, what the overhead costs, and an assessment of where the field actually stands — which is the most professionally useful thing in the course.

Why quantum error correction is hard

Three obstacles that classical coding theory never faces:

1. **No-cloning.** Repetition codes are unavailable. You cannot store three copies.
2. **Continuous errors.** A gate can over-rotate by any angle — not a discrete bit flip.
3. **Measurement destroys.** Reading a qubit to check it collapses the state you are protecting.

All three are circumvented by one idea: **measure the error, not the state.**

Stabilizer measurement

Encode one logical qubit across several physical ones, then measure operators that reveal *whether an error occurred* while revealing nothing about the logical state.

The three-qubit bit-flip code:

$|0\rangle_L = |000\rangle$ $|1\rangle_L = |111\rangle$
 Stabilizers: Z_0Z_1 and Z_1Z_2

Measuring Z_0Z_1 asks "do qubits 0 and 1 agree?" — a parity check. It returns +1 or -1 and says nothing about whether the state is $|000\rangle$, $|111\rangle$, or a superposition. The **syndrome** localizes the error:

Z_0Z_1	Z_1Z_2	diagnosis
+1	+1	no error
-1	+1	qubit 0 flipped
-1	-1	qubit 1 flipped
+1	-1	qubit 2 flipped

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qc = QuantumCircuit(5, 2)          # 3 data + 2 syndrome ancillas
qc.cx(0, 3); qc.cx(1, 3)          # ancilla 3 accumulates parity of  $Z_0Z_1$ 
qc.cx(1, 4); qc.cx(2, 4)          # ancilla 4 accumulates parity of  $Z_1Z_2$ 
qc.measure([3, 4], [0, 1])        # measure ancillas ONLY – data untouched

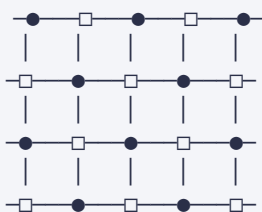
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Continuous errors are digitized by this measurement. An error that rotates a qubit by small angle θ leaves it in a superposition of "no error" and "bit flip." Measuring the syndrome projects onto one or the other — with probability $\sin^2\theta$ you find a full flip and correct it; otherwise you find nothing and the state is restored exactly. The measurement discretizes the error, which is what makes a finite correction set sufficient. This is the single most elegant result in the subject.

Analogy boundary — syndromes as checksums. Parity checks over data blocks, with the syndrome identifying the corrupted position, is genuinely the same mathematics — the surface code is a CSS code built from two classical codes. Two boundaries. First, your checksum reads the data; a stabilizer measurement must extract parity **without** learning any data value, since that would collapse the logical state. Second, quantum codes correct **two independent error types** — bit flips (X) and phase flips (Z) — and correcting one can introduce the other. The three-qubit code above handles only bit flips; Shor's nine-qubit code concatenates two such codes to handle both.

The surface code

The leading architecture. Qubits on a 2-D lattice with only nearest-neighbour interactions — which matters because that is what hardware can build.



- data qubit
- syndrome qubit (X or Z stabilizer)

distance d lattice:
 $2d^2 - 1$ physical qubits
 corrects $\lfloor (d-1)/2 \rfloor$ errors

Below a threshold physical error rate $p_{th} \approx 10^{-3}$, logical error is suppressed **exponentially** in code distance:

$$p_{\text{logical}} \approx (p_{\text{physical}} / p_{th})^{\{(d+1)/2\}}$$

The threshold is the crucial concept. Above it, adding qubits makes things worse. Below it, every increase in d buys exponential improvement for polynomial cost. **Current hardware has crossed the threshold** — Google's 2023 and 2024 results demonstrated that a distance-5 logical qubit outperforms distance-3, confirming the scaling works in practice.

The overhead

p_{physical}	d needed for $p_L = 10^{-15}$	physical qubits per logical
10^{-3}	~ 27	$\sim 1,500$
10^{-4}	~ 13	~ 340

Add magic-state distillation for T gates — each logical T consumes many distilled states, and distillation factories can occupy the majority of a device's qubits.

RSA-2048 in these terms: $\sim 4,100$ logical qubits $\times \sim 5,000$ physical each $\approx 2 \times 10^7$ **physical qubits**, running ~ 8 hours. Against $\sim 10^3$ today. The gap is four orders of magnitude, and it is overhead rather than physics — which is why forecasts vary so widely.

Honest assessment

Demonstrated:

- Quantum error correction below threshold, with logical error decreasing as distance grows.
- Random circuit sampling beyond practical classical simulation (Google 2019, USTC 2020–21) — though classical algorithms have repeatedly narrowed these claims, which is itself informative.
- Small-scale chemistry simulation, useful as validation rather than as new science.

Not demonstrated:

- Any commercially valuable computation faster than the best classical method.
- Factoring a number that classical methods could not factor instantly.
- A variational algorithm outperforming classical optimization on a real problem.

Where advantage is most credible, in order: quantum simulation of quantum systems (Feynman's original argument, and the only case where the mapping is native); then cryptanalysis, once fault tolerance arrives; then optimization, where claims remain unsubstantiated and the shot-count economics of Week 9 are frequently omitted.

Analogy boundary — NISQ as early classical computing. The comparison to 1950s mainframes suggests steady scaling toward general usefulness. It misleads in both directions. Worse: without error correction, NISQ devices cannot simply be scaled — noise compounds exponentially, so a bigger noisy device is not a better one, and there is no Moore's-law path from here to there. Better: the theoretical foundation is far more developed than in 1950. We know what fault tolerance requires, we have proofs of the algorithms, and the threshold has been crossed experimentally. The remaining work is engineering at scale, which is a different kind of hard than "we do not know if this is possible."

Ethics and near-term posture

Cryptographic migration is the concrete near-term obligation, and it is not speculative. Harvest-now-decrypt-later means data requiring 30-year confidentiality needs post-quantum protection today. NIST's standards (ML-KEM, ML-DSA, SLH-DSA) were finalized in 2024; the engineering question is crypto-agility — whether your systems can swap primitives without redesign.

Access and concentration. Quantum computing requires capital at a scale that concentrates capability in a few states and firms. Cloud access mitigates this partially and creates its own dependency.

Overclaiming. The field has a persistent gap between marketing and results. As someone who will be asked to evaluate these claims professionally, the questions worth asking: *Logical or physical qubits? What is the two-qubit gate fidelity? Was the classical baseline the best known algorithm or a convenient one? Was the circuit compiled with knowledge of the answer?*

Key takeaways

- QEC works by measuring stabilizers — error syndromes — without measuring the logical state, circumventing no-cloning and measurement collapse.
- Syndrome measurement **digitizes continuous errors**, which is why a finite correction set suffices.
- Below the $\sim 10^{-3}$ threshold, logical error falls exponentially in code distance. Hardware has crossed this threshold.

- Overhead is the barrier: $\sim 10^3$ – 10^4 physical qubits per logical qubit, putting RSA-2048 at $\sim 10^7$ physical qubits against $\sim 10^3$ today.
- No commercially valuable quantum advantage has been demonstrated. Cryptographic migration is nonetheless a present-tense obligation.

Self-check

Q1 (Objective 8) — Explain how syndrome measurement corrects an error of arbitrary continuous angle using only discrete corrections, and why this does not violate the continuity of quantum evolution.

Q2 (Objective 8) — A vendor announces 1,000 qubits and claims imminent RSA-2048 factoring. Write the assessment you would give a CISO, with the specific questions that determine whether the claim is credible.

Check your answers

A1 — Consider a small rotation error on one qubit of $|\psi\rangle_L = \alpha|000\rangle + \beta|111\rangle$:

$$E = \cos \theta \cdot I + i \sin \theta \cdot X_0$$

$$E|\psi\rangle_L = \cos \theta \cdot |\psi\rangle_L + i \sin \theta \cdot X_0|\psi\rangle_L$$

The state is now a **superposition of "no error" and "bit flip on qubit 0."** The error is continuous — θ can take any value.

Now measure the stabilizers Z_0Z_1 and Z_1Z_2 . The no-error component has syndrome (+1, +1); the flipped component has (−1, +1). These are orthogonal eigenspaces, so the measurement **projects** onto one:

outcome (+1,+1) with probability $\cos^2\theta \rightarrow$ state is exactly $|\psi\rangle_L$, no correction needed
 outcome (−1,+1) with probability $\sin^2\theta \rightarrow$ state is exactly $X_0|\psi\rangle_L$, apply X_0 to fix

Either way the post-measurement state is *exactly* correct — no residual error of size θ remains. The continuous parameter has been converted into a discrete outcome plus a probability.

This does not violate continuity of evolution because **measurement is not unitary evolution**. The unitary part of the dynamics remains perfectly continuous; the projection is the non-unitary step, and it is the same projection postulate from Week 10 used constructively. The deep point: the set of

correctable errors need only span the space of possible errors, not enumerate them. Correcting {I, X, Y, Z} on each qubit suffices to correct *any* single-qubit error, because those four operators form a basis for all 2×2 matrices.

A2 — Assessment: not credible on the stated facts, and no change to migration planning.

The claim fails on arithmetic before any evaluation of the vendor. RSA-2048 requires $\sim 4,100$ *logical* qubits. At current physical error rates, each logical qubit costs $\sim 10^3$ – 10^4 physical qubits under the surface code, giving $\sim 2 \times 10^7$ physical qubits. A 1,000-qubit device is four orders of magnitude short. If the 1,000 are physical — which they invariably are — the device supports at most a handful of logical qubits, and plausibly none at useful distance.

Questions that determine credibility:

1. **Logical or physical qubits?** If unstated, assume physical. A specific logical-qubit count at a stated code distance and logical error rate is the only meaningful figure.
2. **Two-qubit gate fidelity and T2?** Below-threshold error ($< 10^{-3}$) is necessary for any fault-tolerance claim. Ask for randomized benchmarking numbers, not marketing figures.
3. **What has actually been factored, and how?** If the answer is 15, 21, or 35, ask whether the modular exponentiation circuit was general or compiled with knowledge of the factors — the 2013 Smolin result showed most such demonstrations reduce to trivial circuits.
4. **Is there a published resource estimate?** Peer-reviewed estimates (Gidney–Ekerå 2021 is the standard reference) put RSA-2048 at ~ 20 million physical qubits and 8 hours.
5. **What is the classical baseline?** Compare against the best known classical algorithm, not a convenient one.

Recommendation regardless of this vendor's claim: proceed with post-quantum migration on the existing schedule. The justification is harvest-now-decrypt-later, not any particular announcement — traffic captured today is decryptable whenever a capable machine appears, so the migration deadline is set by your data's confidentiality lifetime, not by the hardware roadmap. Prioritize long-lived signatures and archival confidentiality; treat crypto-agility as the durable requirement, since it is the only property that survives being wrong about timing in either direction.

End of course. Final exam: 7–11 December 2026, cumulative.